Day (12)

## * Package in python :-

→ The function concept is used for performing some operation and provides codes reusability within the same program and unable to provide code re-usability across the programs.

→ The modules concept is a collection of variables, functions and classes and we can re-use the code across the programs provided module name and main program present in same folder but unable to provide code-reusability across the folders / drivers / environments.

→ The package concept is a collection of modules.

→ The purpose of packages is that to provide code reusability across the folders / drivers / environments.

→ To deal with the package, we need to the learn the following.
a) create a package
b) reuse the package.

### a) Create a package :-

→ To create a package, we use the following steps.

(I) Create a Folder

(II) place/write an empty python file called `__init__.py` (optional)

(III) place/write the module(s) in the folder where it is considered as package name.

example.

```
bank            ←  packagename
__init__.py     ← - - -  empty python file
simpleint.py    ← - - -  module name
aop.py  - - - -          module name
icici.py  - - -            ,,
welcome.py  - - -          ,,
```

## b) re-use the package

→ To the re-use the modules of the packages across the folders/ drivers/ environments, we have to two approaches. They are:

(I) By using sys.path.append()

(II) By using PYTHONPATH environmental variable Name

### (I) By using sys.path.append()

syntax: $\underbrace{\text{sys.path.append("Absolute path of package")}}$

=> 'sys' is predefined module

=> path is predefined object of list/ variable present in sys module.

=> append() is predefined function present in path and is used for locating the package name of python (specify the absolute path)

e.g).

sys.path.append("D:\\Arif-PYTHON-6PM\\PAckages\\Bank").

or "\" ← single me bhi likh sakte hai

or "/" ← isme me bhi lik sakte hai

### (II) By using PYTHONPATH Environmental variables :-

=> PYTHONPATH is one of the environmental variable.

=> search for environmental variable.

steps for setting

varname : PYTHONPATH.

var value : Address of file.

overall path PYTHONPATH= D:\\Arif-PYthon\\packages\\Bank interview question

| Function | Module | Package |
|---|---|---|
| → Function is a subprogram, which is used to perform certain operation and provides code re-usability | → A module is a collection Global variables, function names and class Names. | → A package is a collection modules. |
| → A function related code can be accessed within same program But not possible to access Across the program. | → The purpose of module is that to re-use the Global variables, Function names and Class names of module either within the same or across the programs provided module and its corresponding main program must present in same Folder But not possible Access across the Folders or environments or networks. | → The purpose of package is that to re-use modules (Global variables, Function Names and Class Names) either within the same Folder or Different or Environments or networks. |

NOTE:

The concept of functions, modules, and packages are called re-usable Techniques in functional programming